OPERATIONS RESEARCH

Dr. Ali Fuat ALKAYA

What is Operations Research (OR)? - 1

- Analyzing the problem of "allocating available resources to various activities" in an organization in a way which is most effective for the organization as a whole.
- Application of scientific methodology to any problem relating to the optimal management of some system.

What is Operations Research (OR)? - 2

- It is the systematic application of quantitative methods, techniques and tools to the analysis of problems involving the operation of systems.
- Operations research emphasizes the basic understanding of the functioning of complex systems of technology and management through mathematical models for the purpose of predicting system behavior or optimizing system performance under economic and technological constraints.

What is Operations Research (OR)? - 3

- "OR is the application of scientific methods, techniques and tools to problems involving the operations of a system so as to provide those in control of the system with optimum solutions to problems."

What is Operations Research (OR)? - 4

- Operations:
  - The activities carried out in an organization.
- Research:
  - The process of observation and testing characterized by the scientific method.

What is Operations Research (OR)? - 5

- OR may simply be viewed as a systematic and analytical approach to decision making or problem solving.
- OR is not an absolute decision-making process, but it is an aid to making good decisions.

What is Operations Research (OR)? - 6

- OR plays an advisory role by presenting to a manager or a decision maker scientifically derived alternatives.
- However, the final decision is always left to the human being who has knowledge that cannot be exactly quantified and who can temper the results of the analysis to arrive at a sensible decision.

What is Operations Research (OR)? - 7

- It involves
  - Analyzing and building mathematical models of complex business situations
  - Solving and refining the mathematical models to gain insight into the business situation

OR and MS

- Sometimes OR is referred to as management science (MS) to better reflect its role as a scientific approach to solving management problems, but it appears that this terminology is more popular with business professionals.

What is the historical background of OR? - 1

- OR as a formal discipline is about 70 years old and its origins may be traced to the latter half of World War II.
- OR methods were first formally defined and used in World War II, in optimal allocation of scarce war equipment and material.
- Programming was a military term at that time meaning planning or scheduling.

What is the historical background of OR? - 2

- Groups of mathematicians, economists and other scientists were formed in England and in US
- These groups tackled a variety of problems such as
  - radar installations,
  - search for enemy submarines,
  - deploy mines in the seas around Japan,
  - determining optimal size of merchant convoy fleets,
  - development of maneuver strategies for ships under attack
  - ...

What is the historical background of OR? - 3

- After World War II
  - Theoretical background of OR was strengthen
  - Industrial firms in England and US attempting to apply it to their operational and managerial problems

What is the historical background of OR? - 4

- Some milestones:
  - Leonid Kantorovich, a Russian mathematician who developed linear programming problems in 1939
  - George Dantzig, who published the simplex method in 1947 (Programming in a linear structure, 1947)
  - Remember, programming was a military term at that time meaning planning or scheduling
  - John von Neumann, who developed the theory of the duality in 1947
- Other significant theoretical developments in 1950-1970 period.

What is the historical background of OR? - 5

- Widespread use and success in practical applications in 1975-1995 period (parallel to the developments in computer technology).
- Today, operations research is a mature, well-developed field with a sophisticated array of techniques that are used routinely to solve problems in a wide range of application areas.

Why apply OR?

- Identify efficient ways of doing business
  - save money
  - increase output/quality
  - optimize use of scarce resources
  - analyze cause-effect relationships

Where has it been applied? - 1

- Just about any kind of problem you can think about: some examples,
  - management/business
  - healthcare
  - banking and finance
  - marketing
  - manufacturing/production
  - design
  - planning

Where has it been applied? - 2

- Just about any kind of problem you can think about: some examples,
  - transportation/airlines
  - scheduling
  - resource allocation
  - chemical/oil production/blending
  - defense/military
  - e-commerce
  - telecommunications

Where has it been applied? - 3

- Burger King (how different cuts of meats should be blended - LP)
- American Airlines (aircraft crew scheduling - ILP)
- LA roadway rebuilding (project scheduling - CPM)
- Finland (planning environmental policy)
- Disneyland/Disneyworld (design plan for parks – Queuing)
- Mrs. Fields' Cookies (when/how much to bake – Forecasting, Inventory Analysis)

Some Success Stories

Company	Problem	Techniques Used	Annual Savings
Hewlett Packard	Designing buffers into production line	Queuing models	$280 million
Taco Bell	Employee scheduling	IP, Forecasting, Simulation	$13 million
Proctor & Gamble	Redesign production & distributon system	Transportation models	$200 million
Delta Airlines	Assigning planes to routes	Integer Programming	$100 million
AT&T	Call center design	Queuing models, Simulation	$750 million
Yellow Freight Systems, Inc.	Design trucking network	Network models, Forecasting, Simulation	$17.3 million
San Francisco Police Dept.	Patrol Scheduling	Linear Programming	$11 million
Bethlehem Steel	Design an Ingot Mold Stripper	Integer Programming	$8 million
North American Van Lines	Assigning loads to drivers	Network modeling	$2.5 million
Citgo Petroleum	Refinery operations & distribution	Linear Programming, Forecasting	$70 million
United Airlines	Scheduling reservation personnel	LP, Queuing, Forecasting	$6 million
Dairyman's Creamery	Optimal production levels	Linear Programming	$48.000
Phillips Petroleum	Equipment replacement	Network modeling	$90.000

What is Decision Making?

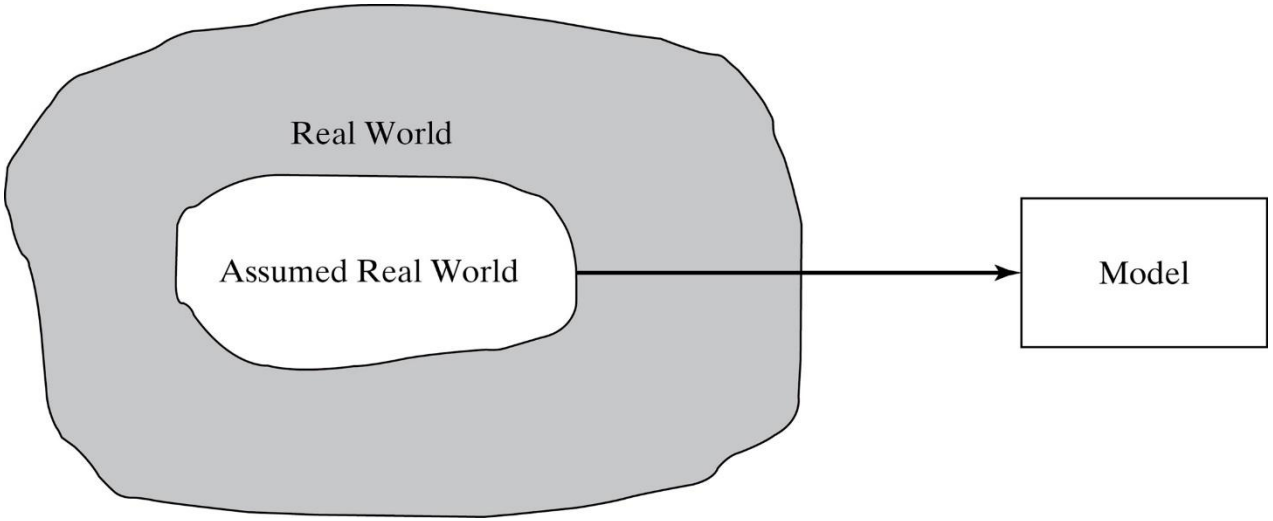
- Decision is to select the best alternative among the others.
- Simple decisions are apparent and easy to find the best alternative.
- Hard decisions are related with complex systems and need more comprehensive analysis of the system.

An example of human factor

- Presence of human element in most decision processes is one of the qualitative factors
- An illustration of the human factor is a version of the widely circulated elevator problem.
  - In response to tenants' slow elevator service in a large office building, the situation was analyzed using a waiting-line model
  - The proposed solution to speed service did not alleviate the problem.
  - A further study revealed that tenant's complaints were more a case of boredom.
  - Problem solved by installing full-length mirrors at elevator entrance.

What is mathematical modeling?

- A model is a representation of an actual object or situation.
- It is an abstraction of reality; it must be less complex yet be representative.



Application Steps of the Operations Research Approach

- Given that OR represents an integrated framework to help making decisions, the next step is to have a clear understanding of this framework so that it can be applied to a generic problem.
  - Problem Definition
  - Data Collection
  - Model Formulation
  - Verification of the Model
  - Solution of the Model
  - Model Validation and Output Analysis
  - Implementation and Monitoring

Problem definition - 1

- A clear definition of the problem contains three broad components:
  - The first is the statement of an unambiguous objective.
  - The second component of problem definition is specification of factors that will affect the objective.
  - The third and final component of problem definition is a specification of the constraints on the system activities

Problem definition - 2

- A clear definition of the problem contains three broad components:
- The first is the statement of an unambiguous objective.
  - Along with a specification of the objective, it is also important to define its scope, that is, to establish limits for the analysis to follow.
  - While a complete system level solution is always desirable, this may often be unrealistic when the system is very large or complex, and in many cases one must then focus on a portion of the system that can be effectively isolated and analyzed.
  - Some examples:
    - to maximize profits over the next quarter from the sales of our products,
    - to minimize the average downtime at work center X,
    - to minimize total production costs at Plant Y, or
    - to minimize the average number of late shipments per month to customers.

Problem definition - 3

- The second component of problem definition is specification of factors that will affect the objective.
  - These factors must further be classified into;
    - Factors that under the control of the decision maker
    - Uncontrollable factors over which he or she has no control.
  - For example, in a production environment, the planned production rates can be controlled, but the actual market demand may be unpredictable.
  - The idea here is to form a comprehensive list of all the alternative actions that can be taken by the decision maker and that will then have an effect on the stated objective.
  - Eventually, the OR approach will search for the particular course of action that optimizes the objective

Problem definition - 4

- The third and final component of problem definition is a specification of the constraints on the system activities
  - setting boundaries for the specific activities that the decision maker may take.
  - As an example, in a production environment the availability of resources may set limits on what levels of production can be achieved.

Data collection

- One of the major driving forces behind the growth of OR has been the rapid growth in computer technology and the concurrent growth in information systems and automated data storage and retrieval.
- This has been a great boon because OR analysts now have ready access to data that was previously hard to obtain.
- Simultaneously, this has also made things difficult, because many companies find themselves in the situation of being data-rich but information-poor. In other words, even though the data is all present somewhere and in some form, extracting useful information from these sources is often very difficult.
  - Data mining
  - Business Analytics

An example:

- Consider a highly simplified instance of a production planning problem where there are two main product lines (X and Y, say) and three major limiting resources (A, B, and C) for which each of the products compete.
- Each product requires varying amounts of each of the resources and the company incurs different costs (labor, raw materials etc.) in making the products and realizes different revenues when they are sold.
- The objective of the OR project is to allocate the resources to the two products in an optimal fashion.

Related data of the example

- To relate data collection to our hypothetical production example, based on variable costs of production and the selling price of each of the products, it might be determined that
  - the profit from selling one X is $10 and one Y is $9.
- It might be determined based on time and work measurements that each X and each Y respectively requires;
  - 7/10 unit and 1 unit of resource A,
  - 1 unit and 2/3 unit of resource B, and
  - 1/10 unit and 1/4 unit of resource C
- Finally, based on prior commitments and historical data on resource availability, it might be determined that in the next month there will be following capacities of the resources available for use in producing the two products;
  - 630 units of resource A,
  - 708 units of resource B, and
  - 135 units of resource C

Model formulation - 1

- A model may be defined formally as a selective abstraction of reality.
- This definition implies that modeling is the process of capturing selected characteristics of a system or a process and then combining these into an abstract representation of the original.
- The main idea here is that it is usually far easier to analyze a simplified model than it is to analyze the original system, and as long as the model is a reasonably accurate representation, conclusions drawn from such an analysis may be validly extrapolated back to the original system.

Model formulation - 2

- The key point to remember is that most often there is a natural trade-off between the accuracy of a model and its tractability.
- At the one extreme, it may be possible to build a very comprehensive, detailed, and exact model of the current system. However, the model may be useless from an analytical perspective since its construction may be extremely time-consuming and its complexity precludes any meaningful analysis.
- At the other extreme, one could build a less comprehensive model with a lot of simplifying assumptions so that it can be analyzed easily. However, the danger here is that the model may be so lacking in accuracy that extrapolating results from the analysis back to the original system could cause serious errors.

Model formulation - 3

- There may be no single correct way to build a model.
- OR is an art because the success of all the phases that precede and succeed the solution of the mathematical model depends largely on the creativity and experience of the operations research team.

Verification of the model

- The developed model must be verified to proof its correctness
- Determine whether the model is an accurate representation of the intended model of the real system.
- Verification is the building of model right

Solution of the model

- Model solution is the simplest of all OR phases because it entails the use of well-defined optimization algorithms
- An important aspect of the model solution phase is sensitivity analysis
- Sensitivity analysis deals with obtaining additional information about the behavior of the optimum solution when the model undergoes some parameter changes
- Sensitivity analysis is particularly needed when the parameters of the model cannot be estimated accurately. In this case, it is important to study the behavior of the optimum solution in the neighborhood of the estimated parameters

Validation and output analysis

- The model must predict adequately the behavior of the system under study
- Validation means building the right model
- A common method for checking the validity of a model is to compare its output with historical output data
- The model is valid if, under similar input conditions, it produces past performance
- If the proposed model is representing a new (nonexisting) system, no historical data would be available to make the comparison. In such cases, the use of simulation as an independent tool for verifying the output of the mathematical model generally is resorted

Implementation and monitoring

- Implementation of the solution of a validated model involves the translation of the results into operating instructions issued in understandable form to the individuals who will administer the recommended system

Mathematical Models

- It has traditionally been most commonly identified with OR.
- In this type of model one captures the characteristics of a system or process through a set of mathematical relationships.
- Mathematical models allow us to find optimum solution and to show the relationships between elements of system analytically

Basic elements of mathematical models

- Objective function/functions
- Decision variables
- Constraint functions

Objective function

- Objective function represents considered performance of system in terms of decision variables
- We need to maximize or minimize the objective function according to the purpose
- Objective function must be maximized or minimized according to the considered performance type
  - Profit or cost level respect to amount of manufactured products
  - Travel time according to assigned buses on a route
  - Produced amount of a chemical material
  - Utilized time of an airport according to the scheduled departures, …

Decision variables

- Decision variables can be controlled by the system (or decision maker) and their values effects the performance of the system
- Batch sizes of the products
- Number of buses on a route
- Amount of a chemical material in a chemical process
- Scheduling of airplanes' departures,….

Constraint functions

- Constraints show the limitations of system
- Constraints restrict the levels of decision variables directly or indirectly
- Constraint functions should be represented in terms of decision variables
  - Production capacity of a manufacturing line
  - Maximum allowed time to waiting in a bus stop
  - Process time of a chemical reaction
  - Land requirements to depart or arrive of airplanes for an airport, …

Remember the example

- Decision variables:
  - Number of product X
  - Number of product Y
- Constraints:
  - Resources A, B, and C
- Objective function:
  - Total profit from selling X and Y

Mathematical model of the example

- Total profit =10X + 9Y (objective function)
- Constraint functions
  - (7/10)X + Y ≤ 630 (for resource A)
  - X + (2/3)Y ≤ 708 (for resource B)
  - (1/10)X + (1/4)Y ≤ 135 (for resource C)

Operations research techniques to solve different mathematical models

- Linear Programming
- Goal Programming
- Network Programming
- Integer Programming
- Dynamic Programming
- Nonlinear programming
- Stochastic programming

What is optimization?

- Optimization is to find the best levels of decision variables to maximize (or minimize) considered objective function satisfying all the system constraints.
- Optimized level of objective function is unique but there can be alternative combinations of levels of the decision variables which yield the same optimum objective value.

Linear Programming (LP)

- LP applies to optimization models in which the objective and constraint functions are strictly linear.
- LP technique is used in wide range of applications, including agriculture, industry, transportation, economics, health systems, and the military.
- There are also efficient computational algorithms to solve LP problems with thousands of constraints and variables.
- LP also forms the backbone of the solution algorithms for other OR models, including integer, stochastic, and nonlinear programming.

Assumptions of LP

- Proportionality
- Additivity

} Linearity

Proportionality

- Proportionality requires the contribution of each decision variable in the objective function and its requirements in the constraint functions to be directly proportional to the value of the variable.

Additivity

- Additivity stipulates that the total contribution of all the variables in the objective function and their requirements in the constraint functions are the direct sum of the individual contribution or requirements of each variable.

- We can formulate mathematical model for this general problem of allocating resources to activities. This model is to select the values for x_1, x_2, .., x_n so as to

$$\text{Maximize} \qquad Z = c_1x_1 + c_2x_2 + \cdots + c_nx_n,$$

subject to the restrictions

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &\le b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &\le b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &\le b_m, \end{aligned}$$

$$x_1, x_2, \cdots, x_n \ge 0$$

- $[c_1, c_2, .., c_n]$: vector of objective function coefficients
- $[b_1, b_2, .., b_m]$: vector of right-hand side coefficients (or capacities)
- $\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$: matrix of constraints' coefficients

Developing LP models

- We will experience on development of LP models before learning of solution techniques of them
- One of the simple forms of LP models includes problems with 2 decision variables
- Increasing the number of decision variables increases also both dimension of problem and complexity of the problem
- However, LP problems with thousands of decision variables and constraints can be solved efficiently (in polynomial time with the problem dimension)